WHERE IoT POWERS SMART SUPPLY CHAINS

VIACHAIN™
AN ORBCOMM COMPANY



CT 5300
Hardware Guide

GJD148, Version 02
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1 PRODUCT OVERVIEW

The CT 5300 (model number CT5300) is a low cost, versatile tracking and monitoring device capable of cellular communications over various cellular networks using multi-band radios. The CT 5300 interfaces with wireless sensors and integrated, internal sensors to monitor the status and condition of the asset. The CT 5300 also includes an integrated cargo camera enabling the device to detect the presence of cargo within the tracked asset.

Figure 1: CT 5300



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2 SPECIFICATIONS

2.1 Temperature

Parameter	Value
Operating Temperature Range (excludes battery charging)	-40°C to +70°C (-40°F to +158°F)
Storage Temperature Range	-40°C to +60°C (-40°F to +140°F)
Battery Charging	-20°C to +60°C (-4°F to +140°F)

2.2 Electrical

2.2.1 Maximum Input Power

The CT 5300 is powered through the internal battery that is recharged using an integrated solar panel.

Parameter	Value
Battery Charge	2.5 W

2.2.2 Power Consumption

Operation Mode	Current[1]
Sleep + Power Save	150 µA
Cellular Transmit (pulsed)	1000 mA
Cellular Receive	45 mA
Applications Processing	25 mA
Camera Imaging with Flash	1159 mA

2.3 RF Specifications

2.4 GNSS Module

The major constellations include GPS, GLONASS, BeiDou B1I, and Galileo.

Parameter	GPS	GLONASS	BeiDou B1I	Galileo
Time to First Fix				
Cold Start	29 s	27 s	30 s	41 s
Hot Start	1 s	1 s	1 s	1 s
AssistNow Online	1 s	1 s	1 s	5 s

[1] Based on a nominal battery voltage of 7.4 V.

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Parameter	GPS	GLONASS	BeiDou B1I	Galileo
Sensitivity				
Tracking and Navigation	-167 dBm	-166 dBm	-160 dBm	-161 dBm
Cold Start	-148 dBm	-147 dBm	-146 dBm	-141 dBm
Hot Start	-159 dBm	-159 dBm	-159 dBm	-155 dBm
Accuracy				
Horizontal Position (CEP) 1	1.5 m	4.0 m	2.0 m	3.0 m
Velocity	0.05 m/s			
Heading	0.3 degrees			

2.5 Battery

The CT 5300 uses an internal solar rechargeable lithium-ion battery.

The battery characteristics are:

Parameter	Min.	Typical	Max.	Units
Nominal Voltage	-	7.4	-	V
Capacity	5000	-	-	mAh
Current Output	-	-	7.5	A
Over Discharge Limit	-	5.5	-	V

2.5.1 Battery Voltage Measurement

The device can measure the input voltage over the 5 to 8.4 V range.

2.5.2 Battery Controller

Discharge is controlled so that battery discharge is not allowed over the battery pack maximum discharge temperature.

The battery controller electrical specifications are:

Parameter	Min.	Max.	Units
Fast Charge Current	-	239.6	mA
Fast Charge Temperature	0/32	60/140	°C/°F
Slow Charge Current	-	98.5	mA
Slow Charge Temperature	-20/-4	0/32	°C/°F
Float Voltage	8.0	8.4	V
Charge Time	-	20.9	Hours
Discharge Temperature	-	75/167	°C/°F

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2.6 Memory

2.6.1 External Flash Memory

The camera sensor includes 128 Mbit of non-volatile flash memory.

2.6.2 External Non-volatile Data Memory

The camera sensor includes a second 1 Gbit non-volatile flash memory.

2.7 IR Illuminator

The camera includes two (2) IR illuminators.

The IR illuminator specifications are:

Parameter	Value
Power (per emitter)	2.5 W
Wavelength	850 nm

2.8 Camera Lens

The camera lens can view the entire length of the container with a single camera. It has a 180° field of view.

2.9 Sensors

2.9.1 Temperature Sensor

Parameter	Value
Range	-40 to +85°C (-40 to +185°F)
Accuracy (typical)	±2°C (±3.6°F)

2.9.2 Magnet Switch

The device has a magnet switch for local user interaction.

2.9.3 Tamper Detection

The device includes a second magnetic switch to detect device removal after installation.

2.9.4 Accelerometer

The CT 5300 includes a primary accelerometer to detect motion and a secondary accelerometer for more advanced features such as advanced motion (to trigger a report) and shock detection.

The primary and secondary accelerometer specifications are:

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Parameter	Condition	Min.	Typ.	Max.	Units
Acceleration Range	software selectable	-	±2	-	g
		-	±4	-	g
		-	±8	-	g
		-	±16	-	g
Sensitivity	2 g	-	256	-	LSB/g
	4 g	-	128	-	LSB/g
	8 g	-	64	-	LSB/g
	16 g	-	32	-	LSB/g
Bandwidth Filtering	Selectable via digital interface	40	-	400	Hz

2.10 BLE

The BLE system allows the CT 5300 to communicate as a peripheral with a VIACHAIN device, mobile phone or other BLE enabled host for configuration and debug logging.

The BLE characteristics are:

Parameter	Min.	Typical	Max.	Units
Frequency	2402	-	2480	MHz
BLE Version Compliance	-	5	-	-
Receive Sensitivity	-	-95	-	dBm
Transmit Power	-20	-	8	dBm
Available RAM	-	256	-	kB
Available Flash	-	1	-	MB
Frequency Band	2.4			GHz

2.11 Cellular Radio

The device is capable of global cellular communications over LTE, 2G, and 3G networks.

The cellular radio specifications are:

Parameter	Minimum Specification
LTE Category	Category 1
LTE Bands	1, 2, 3, 4, 5, 7, 8, 12, 13, 18, 19, 20, 25, 26, 28, 34, 38, 39,40, 41, and 66
UMTS/HSPA+ Bands	1, 2, 4, 5, 6, 8, 19
GSM Bands	GSM850/EGSM900/DCS1800/PCS1900

2.12 LED

The CT 5300 has a single tricolor LED to indicate general health of the device.

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2.13 Mechanical

Parameter	Description
Weight with rivet bracket without bracket	 1.4 kg (3.0 lb) 0.7 kg (1.5 lb)

Figure 2: CT 5300 with Rivet Bracket Top View

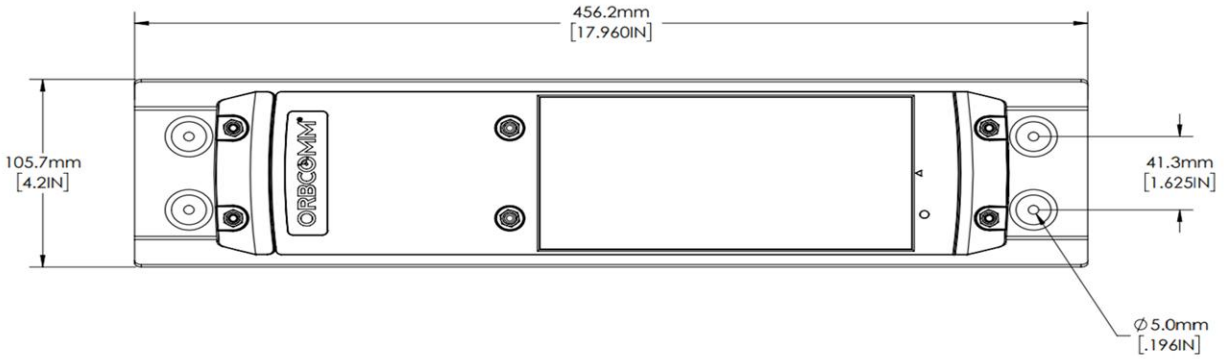


Figure 3: CT 5300 with Rivet Bracket Bottom View

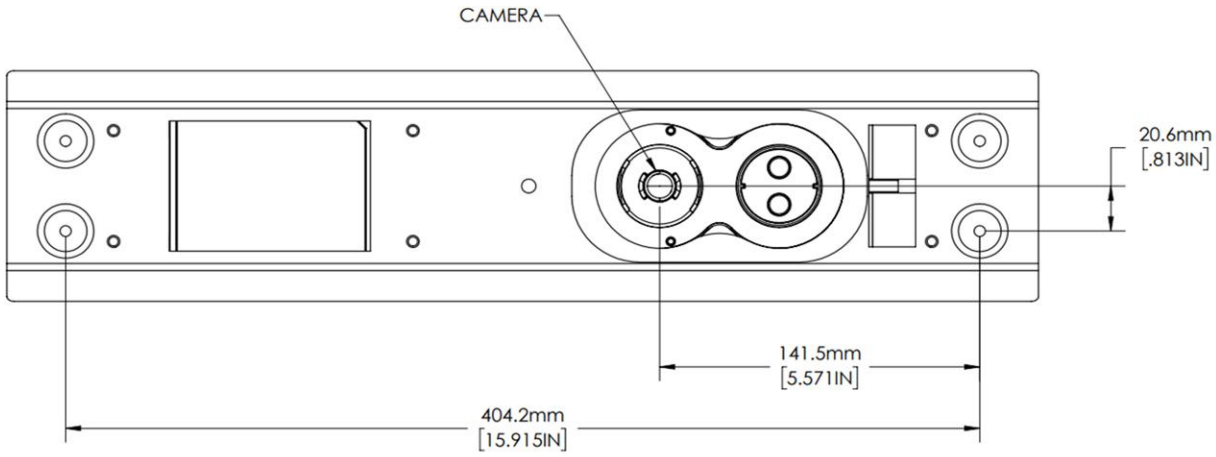
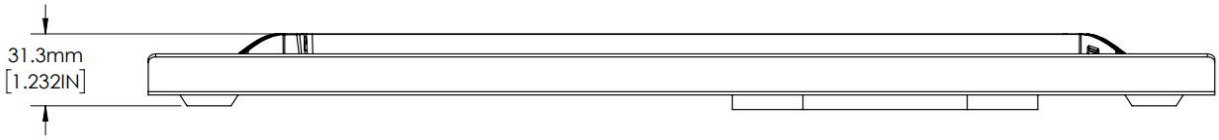


Figure 4: CT 5300 with Rivet Bracket Side View



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3 COMPLIANCE

ISED (Canada)

- IC compliance statement

  This device complies with Industry Canada license-exempt RSS standard(s). Operation is subject to the following two conditions: (1) This device may not cause interference, and (2) this device must accept any interference, including interference that may cause undesired operation of the device.

  Le present appareil est conforme aux CNR d'Industrie Canada applicable aux appareils radio exempts de licence. L'exploitation est autorisée aux deux conditions suivantes: (1) l'appareil ne doit pas produire de brouillage, et (2) l'utilisateur de l'appareil doit accepter tout brouillage radioélectrique subi, meme si le brouillage est susceptible d'en compromettre le fonctionnement.

FCC (USA)

- FCC compliance statement

  This device complies with part 15 of the FCC Rules. Operation is subject to the following two conditions: (1) This device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation. “

CE MARK (Europe)

- RED 2014/53/EU
- EU Declaration of Conformity

  Hereby, ORBCOMM Inc. declares that the radio equipment types listed in this document comply with Directive 2014/53/EU.

  The full text of the EU declaration of conformity is available from http://www2.orbcomm.com/eudoc.

RoHS

- Restriction of Hazardous Substances (RoHS) [1]

REACH

WARNING:

- The minimum 20 cm (8 in.) separation distance from the device is required for RF exposure safety for all persons.
- Changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.
- This device contains licence-exempt transmitter(s)/receiver(s) that comply with Innovation, Science and Economic Development Canada’s licence-exempt RSS(s). Operation is subject to the following two conditions:(1) This device may not cause interference.(2) This device must accept any interference, including interference that may cause undesired operation of the device.

[1] European Union's (EU) Directive 2002/95/EEC "Restriction of Hazardous Substances" (RoHS) in Electronic and Electrical Equipment.

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- L'émetteur/récepteur exempt de licence contenu dans le présent appareil est conforme aux CNR d'Innovation, Sciences et Développement économique Canada applicables aux appareils radio exempts de licence. L'exploitation est autorisée aux deux conditions suivantes :

  1) L'appareil ne doit pas produire de brouillage;

  2) L'appareil doit accepter tout brouillage radioélectrique subi, même si le brouillage est susceptible d'en compromettre le fonctionnement.

Ingress Protection

- CT 5300: IP67

Environmental

Parameter	Description
Vibration	The device meets all its specifications during exposure to random vehicular vibration levels per AAR-S-9401, section 3.2.4.2.
Humidity	The device meets all its specifications during exposure to 90% relative humidity at +85°C (185°F), per the test methodology of SAE J1455, section 4.2.3 (8-hour humidity cycle per figure 4a)
Mechanical Shock	The device meets all its specifications after exposure to positive and negative saw tooth shock pulses with peaks of 40 G and durations of 11 ms as specified in MIL-STD-810H, section 516.8, Procedure 1.
Thermal Shock	The device meets all of its specifications after a thermal shock test as detailed in SAE J1455, section 4.1.3.2
Handling Drop	The device meets all its specifications after a handling drop test as specified in SAE J1455, section 4.11.3.1.
Ingress - Dust and Sand	The device meets all of its specifications after a dust and sand test as detailed in IEC 60529 13.4 and 13.6.2 (IP6x).
Ingress - Water	The device meets all of its specifications after an ingress test per IEC 60529 14.2.7 (IPx7).
Salt Spray Atmosphere	The device meets all of its specifications after a salt spray test as detailed in SAE J1455, section 4.3.3.1.
Steam Cleaning and Pressure Washing	The device meets all its specifications after a steam cleaning and pressure wash test as detailed in ISO 20653 section 8.4.2.
Exposure to Chemicals and Oils	The device meets all of its specifications after a light to moderate splash test as detailed in SAE J1455 section 4.4.3.2, for the relevant chemicals.
Fungus	The device meets all of its specifications after a fungus test as detailed in SAE J1455, section 4.6.3.
ESD (enclosure)	The device meets all its specifications after exposure of the enclosure to 6 kV ESD contact discharge per IEC 61000-4-2, level 3.

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